



Filter Wizard

Filter Wizard Design

Created on 05/31/2025



Filter Wizard Design Report

Filter Requirements for Low-Pass, 4th order Butterworth

Specifications: Optimize: Power; +Vs: 5; -Vs: -5

Gain: 0 dB

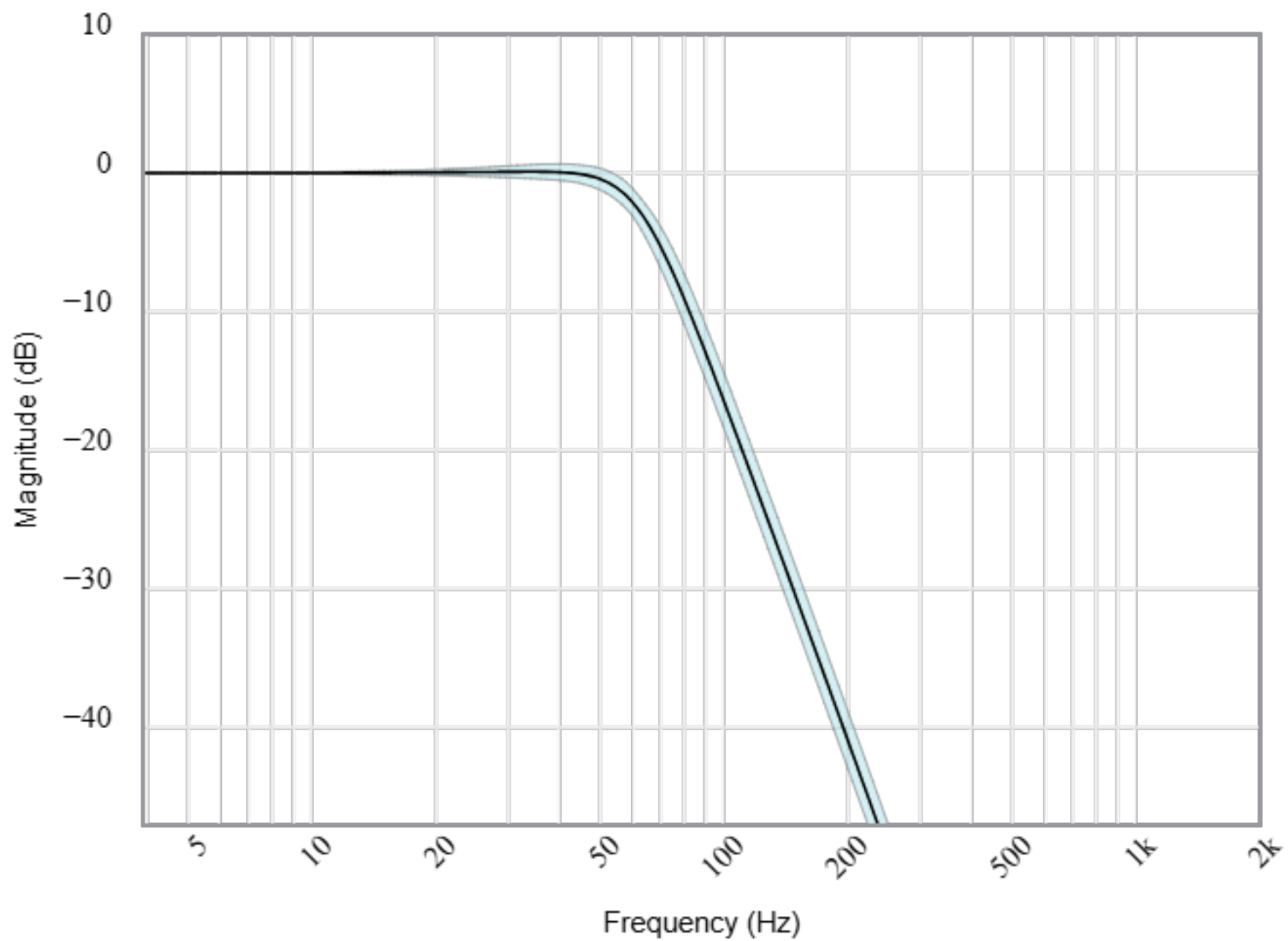
Passband: -0.1dB at 40Hz

Stopband: -27dB at 200Hz

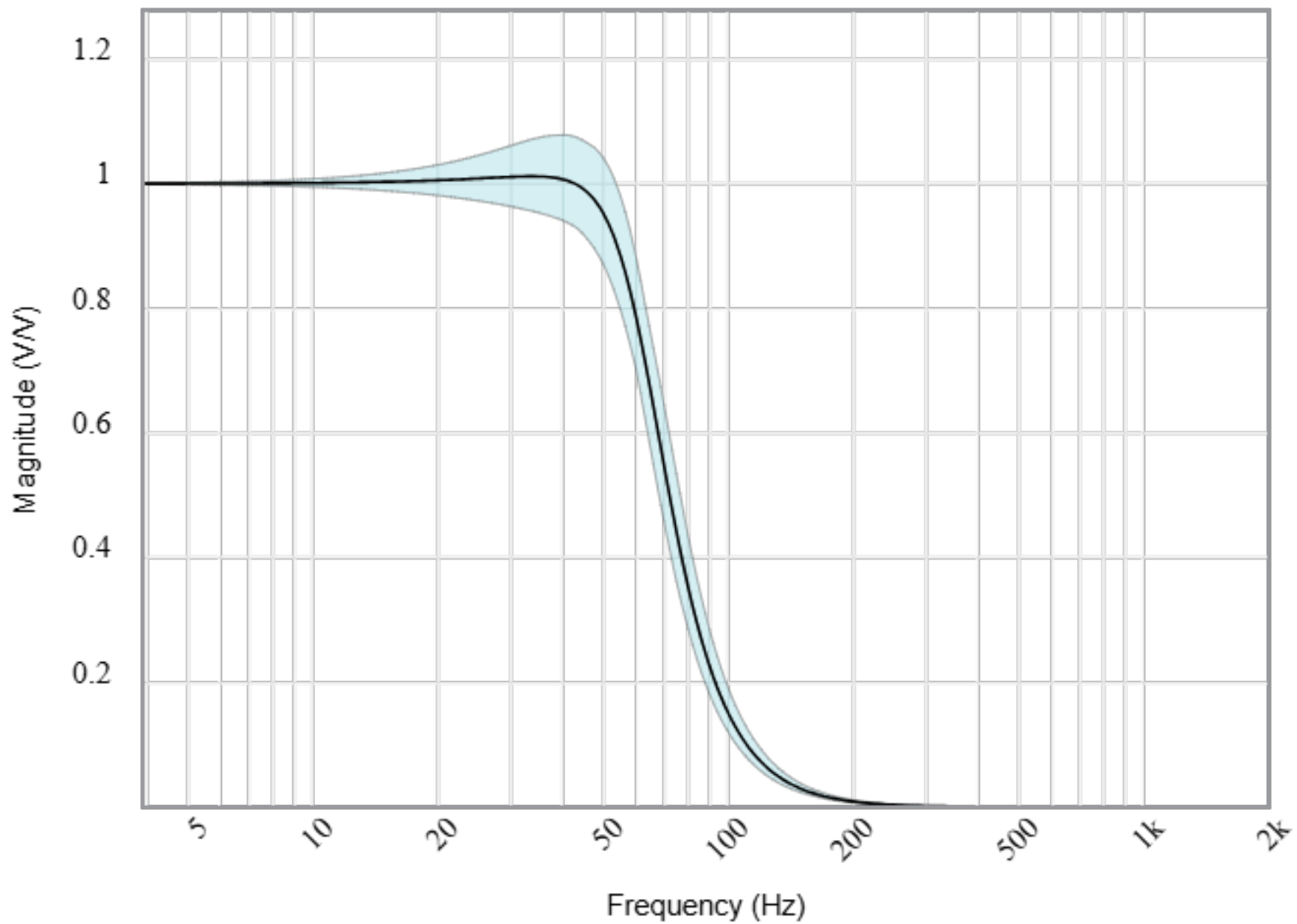
Component Tolerances: Capacitor = 5%; Resistor = 1%; Inductor = 5%; Op Amp GBW = 20%

BOM: refer to BOM.csv file

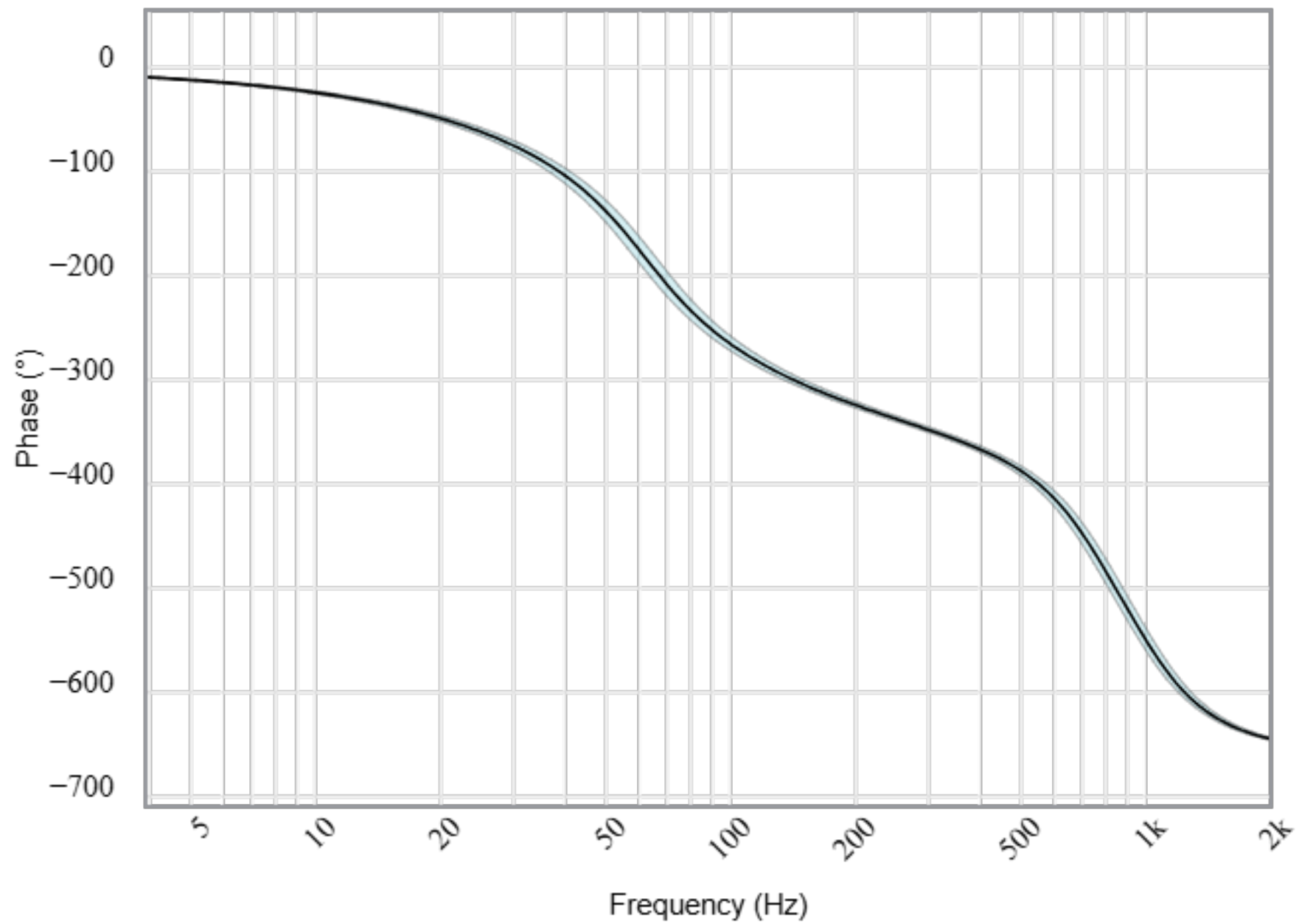
Magnitude(dB)



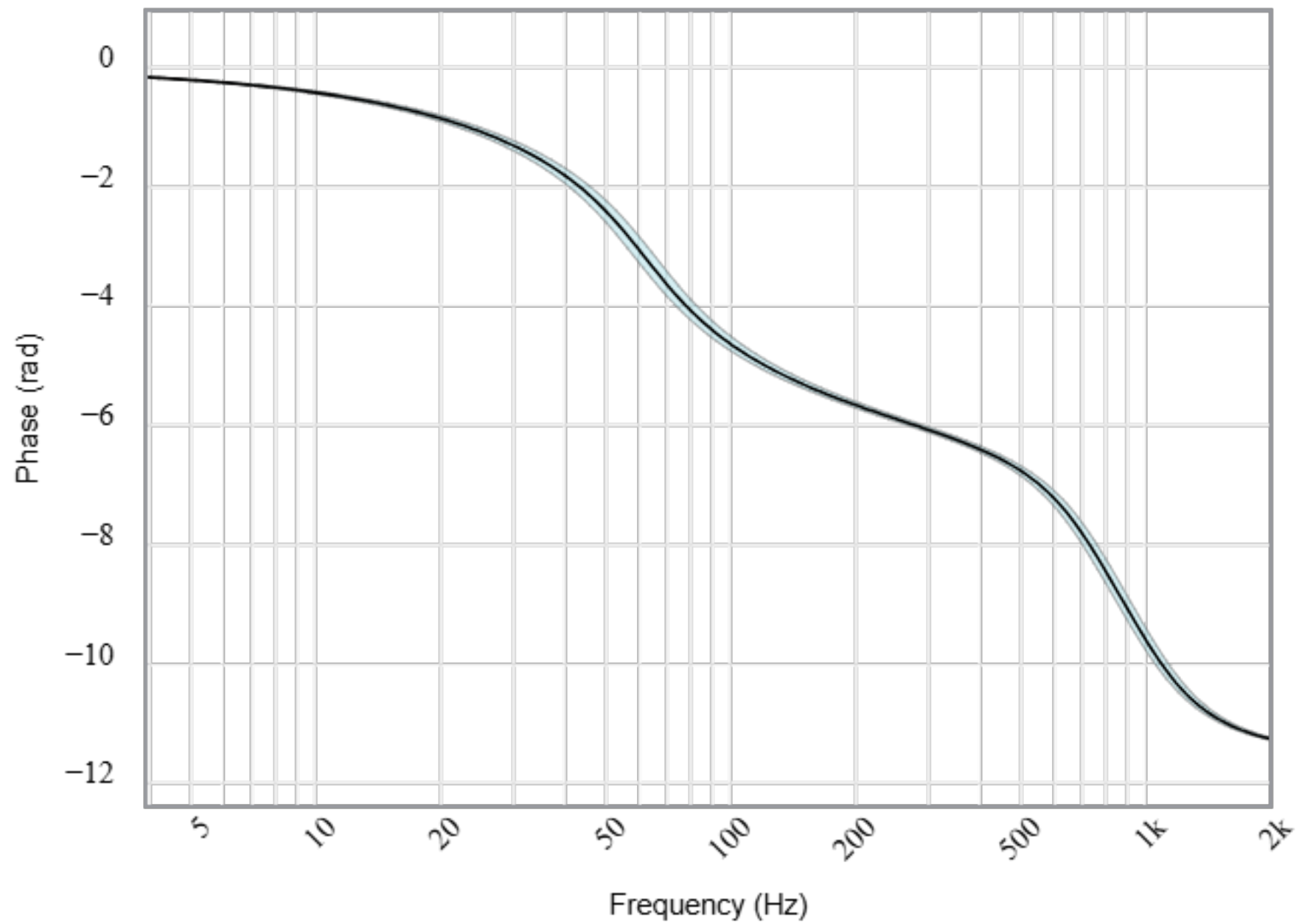
Magnitude(Volts per Volt)



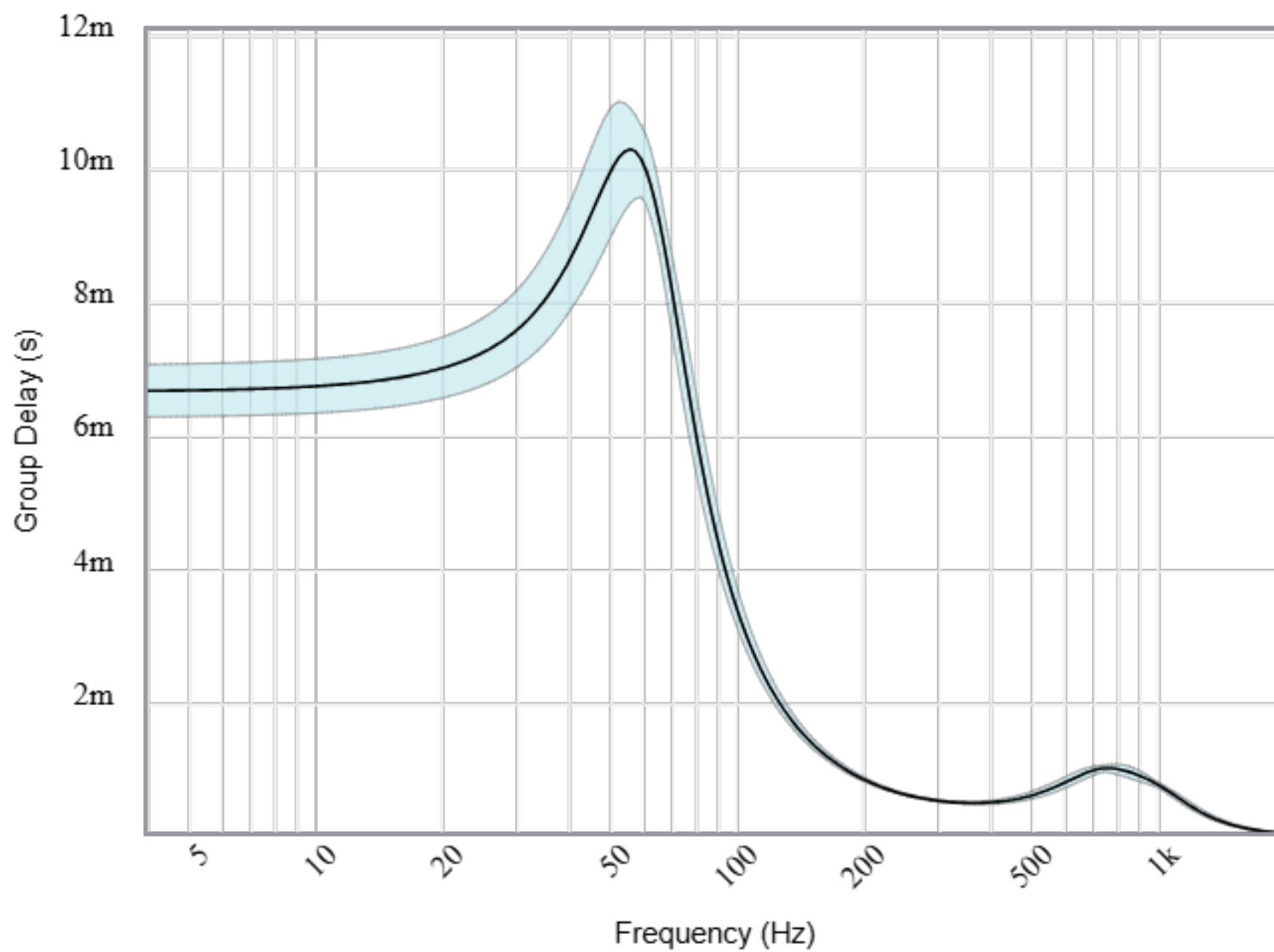
Phase(degrees)



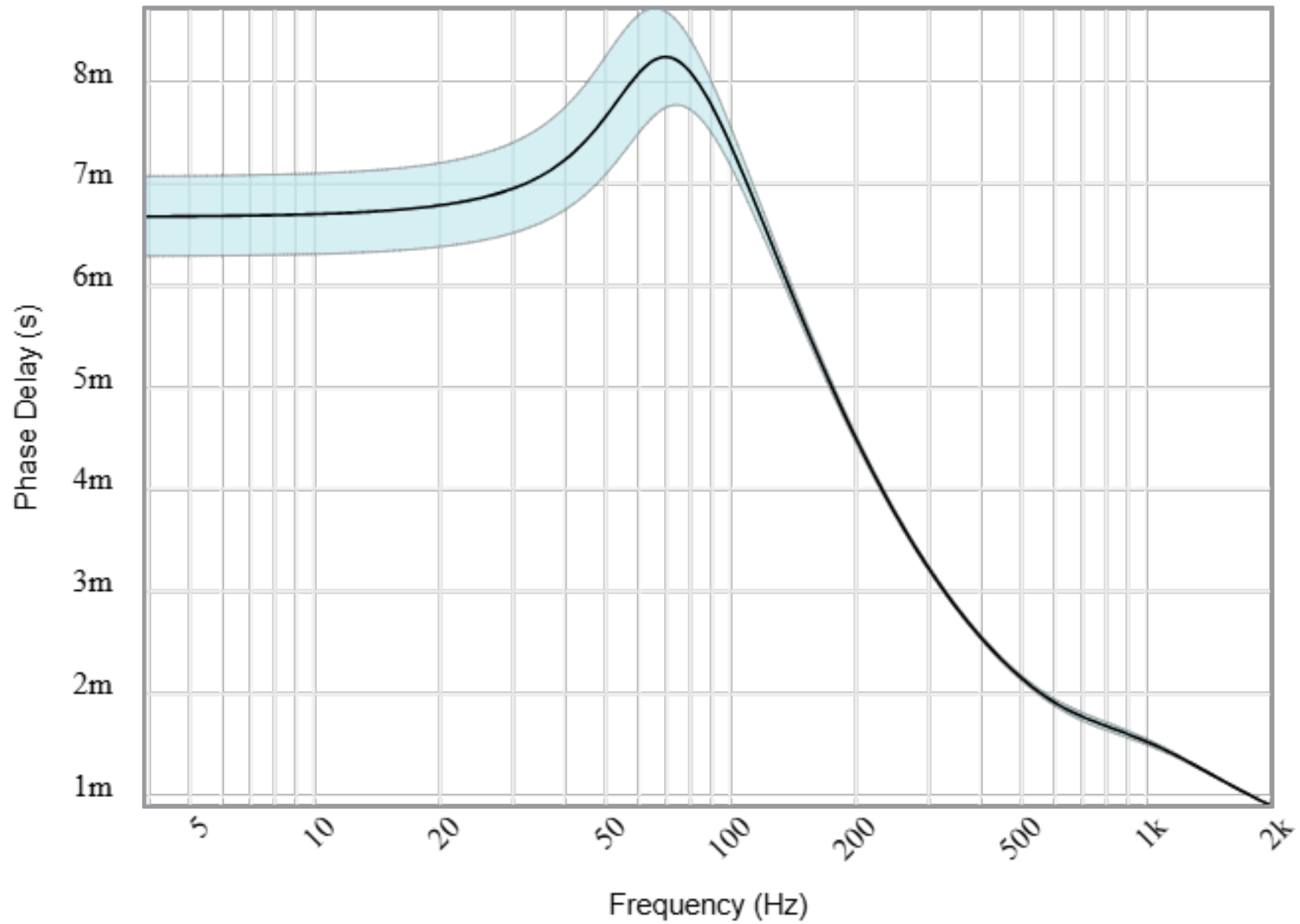
Phase(radians)



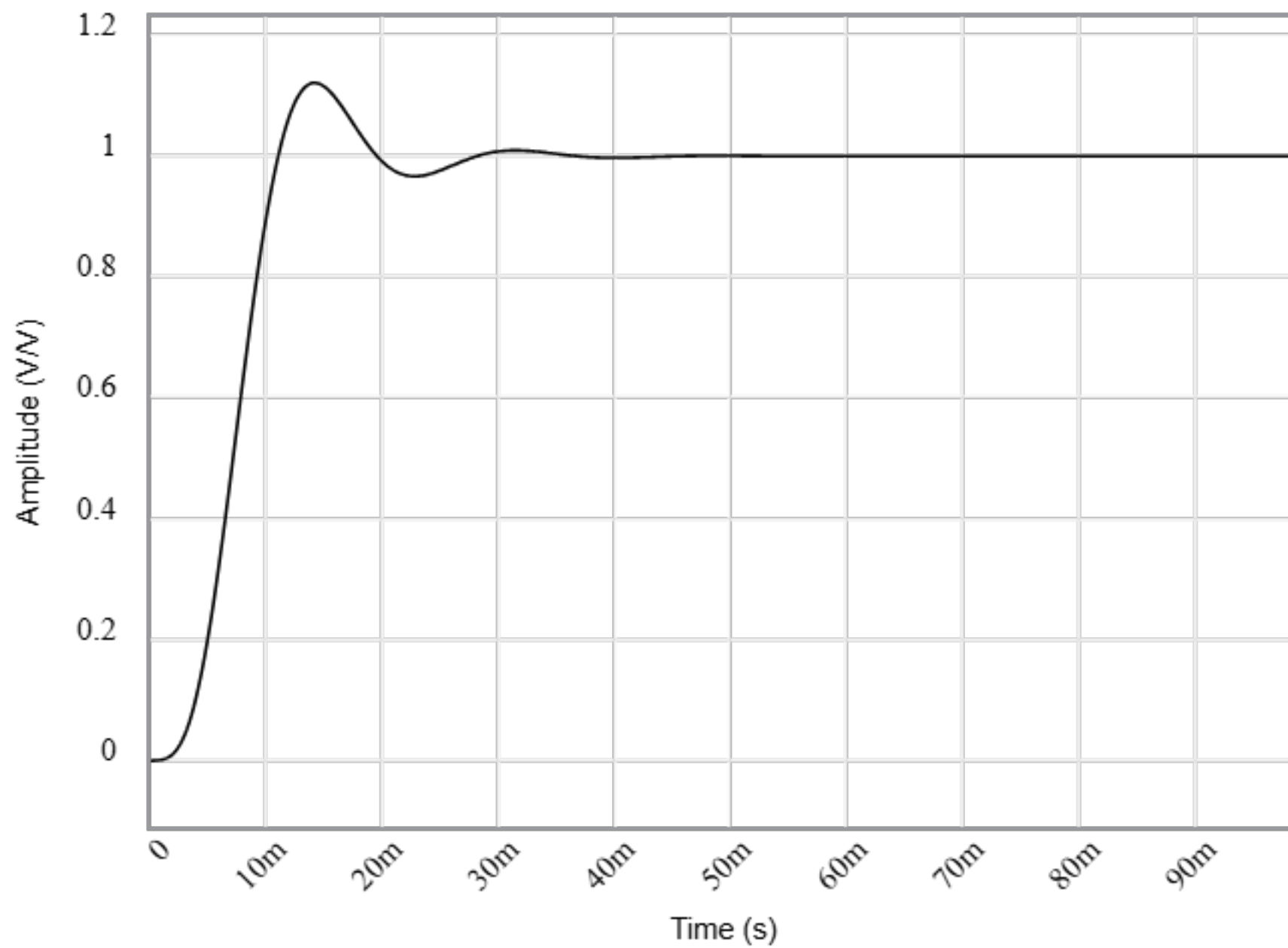
Group Delay



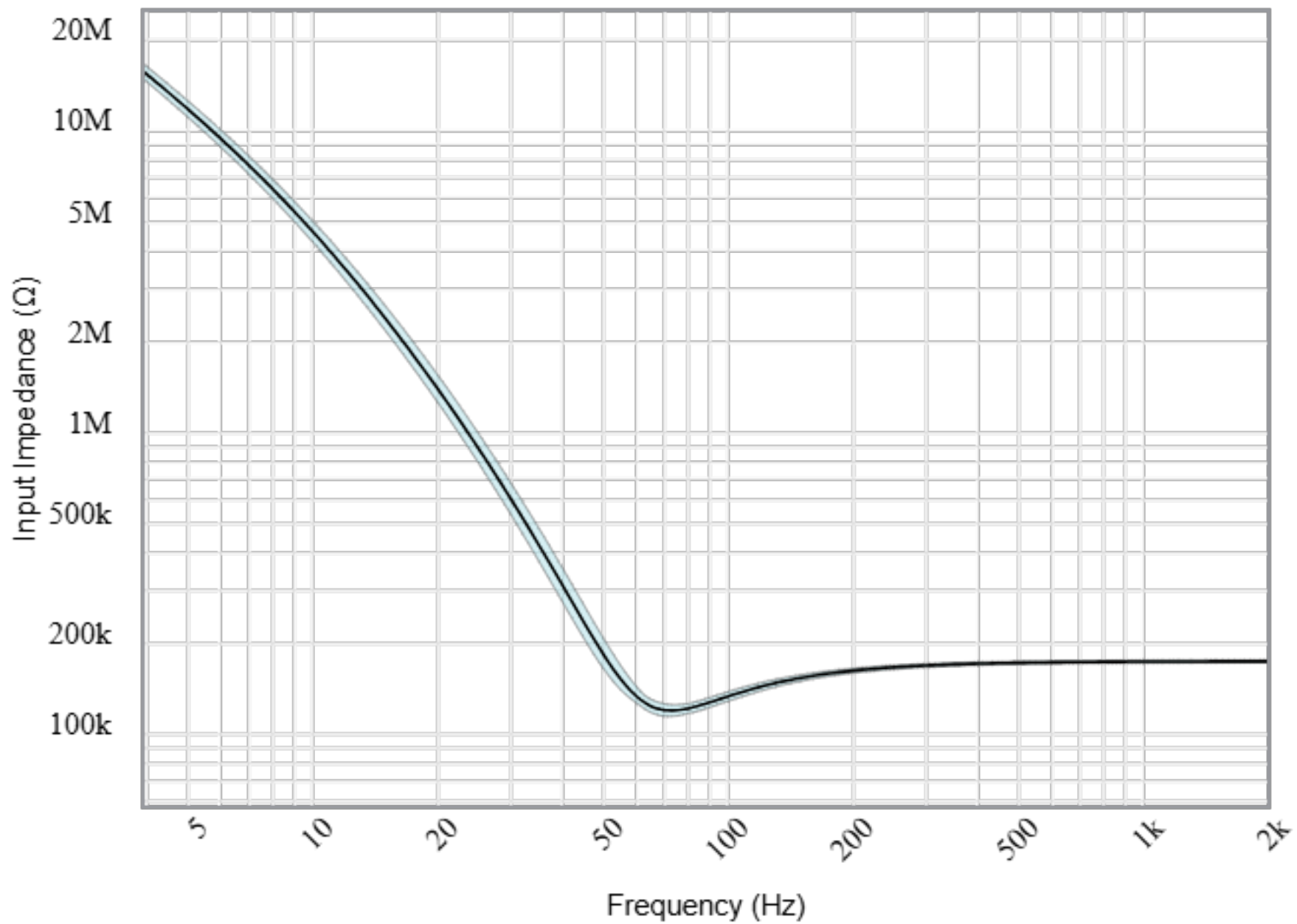
Phase Delay



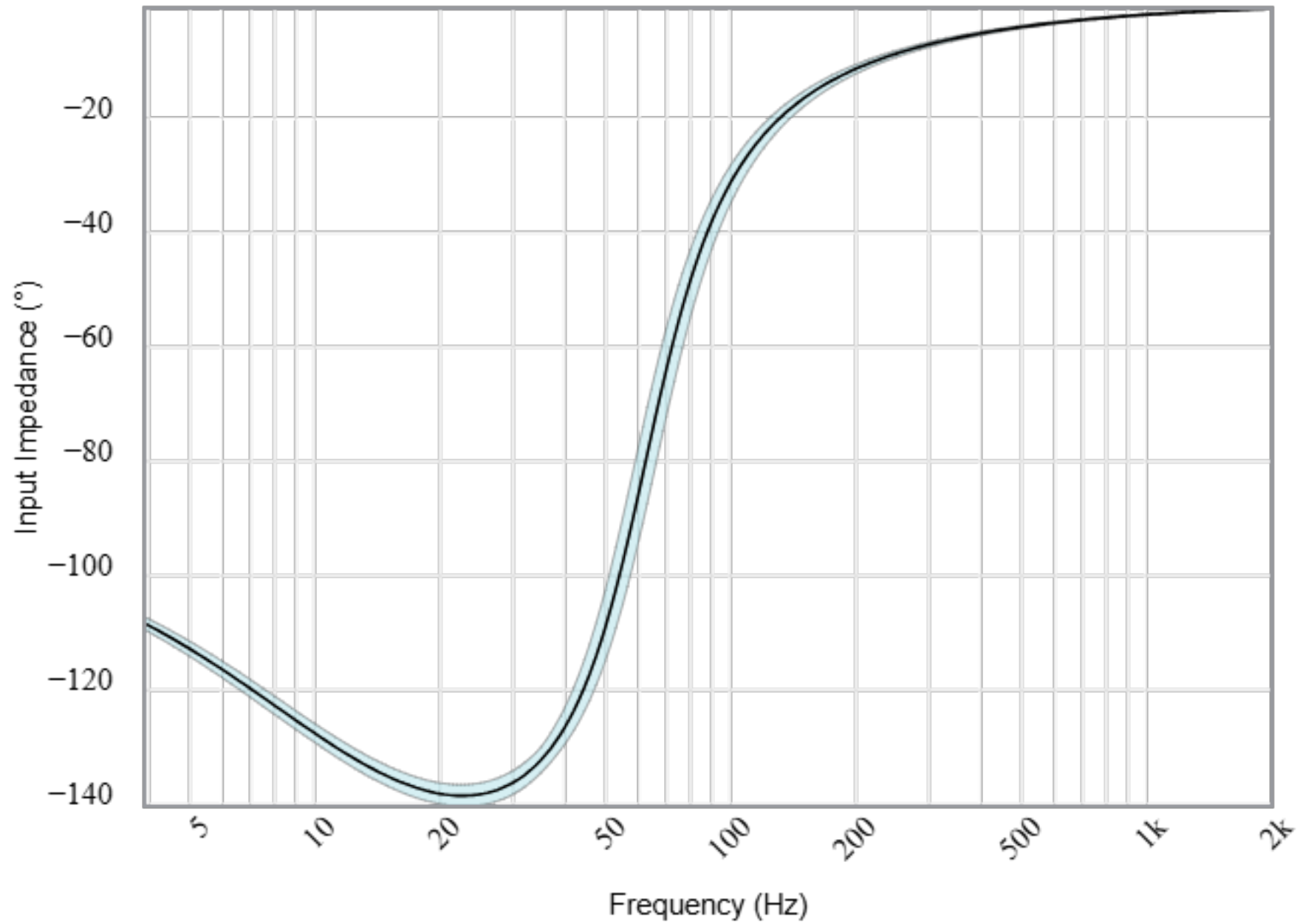
Step Response



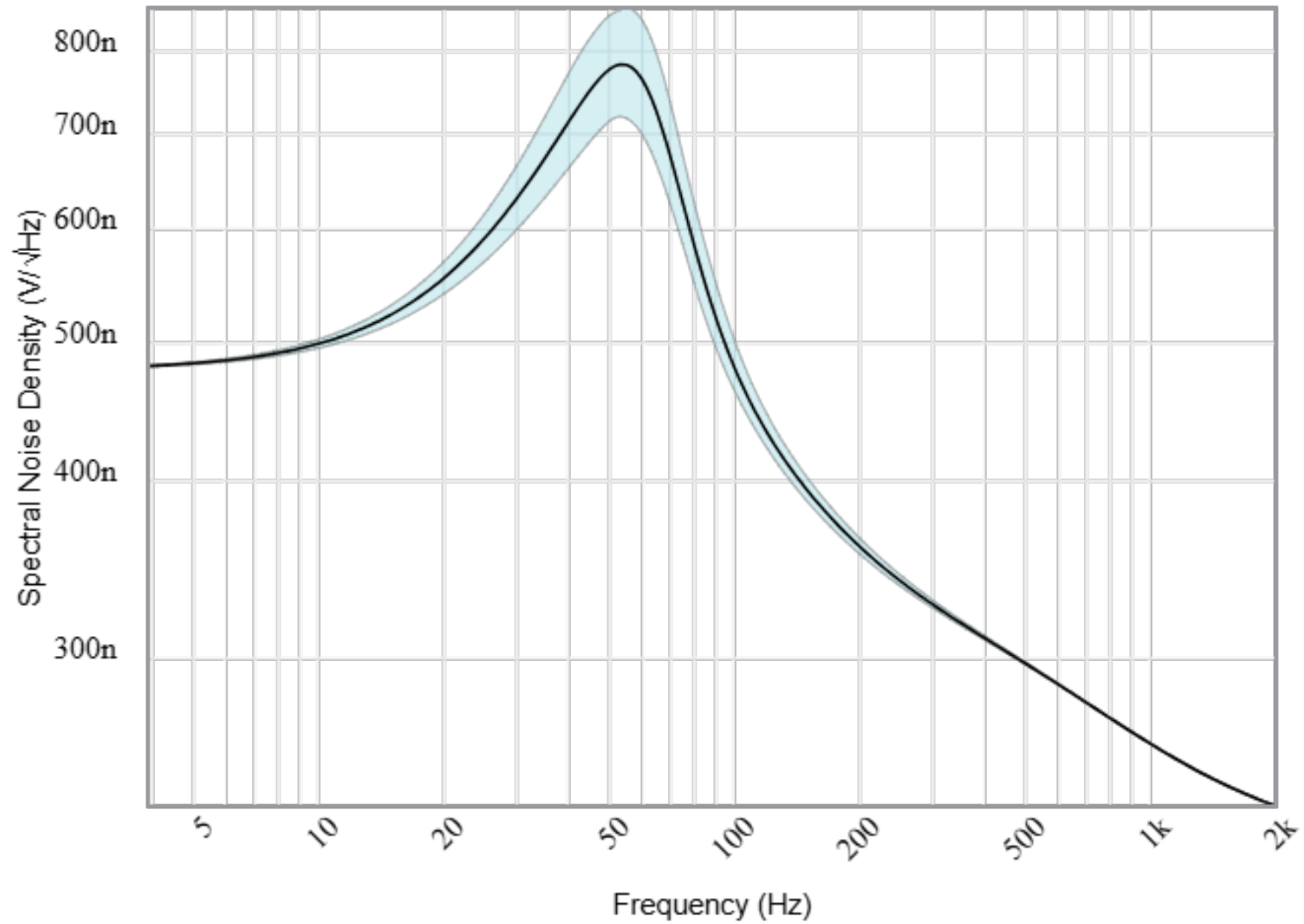
Input Impedance Magnitude



Input Impedance Phase



Noise



Stages

Your filter requires 2 op amp stage(s) with the following characteristics



2nd order
Low-Pass
Sallen Key

Target

Simulated

Gain (V/V):

1

0.999 to 0.999

f_p (Hz):

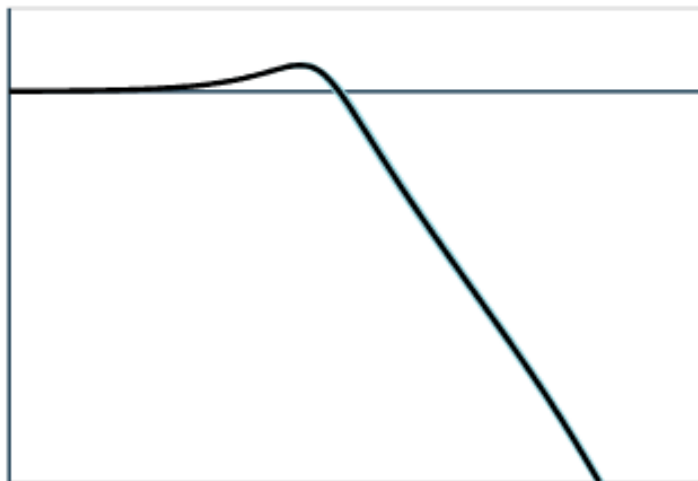
64

58.6 to 65.9

Q:

1.31

1.27 to 1.41



2nd order
Low-Pass
Sallen Key

Target

Simulated

1

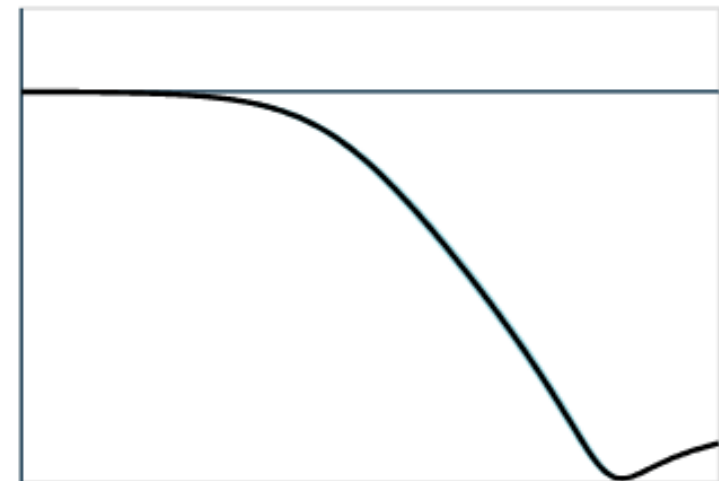
0.999 to 1

64

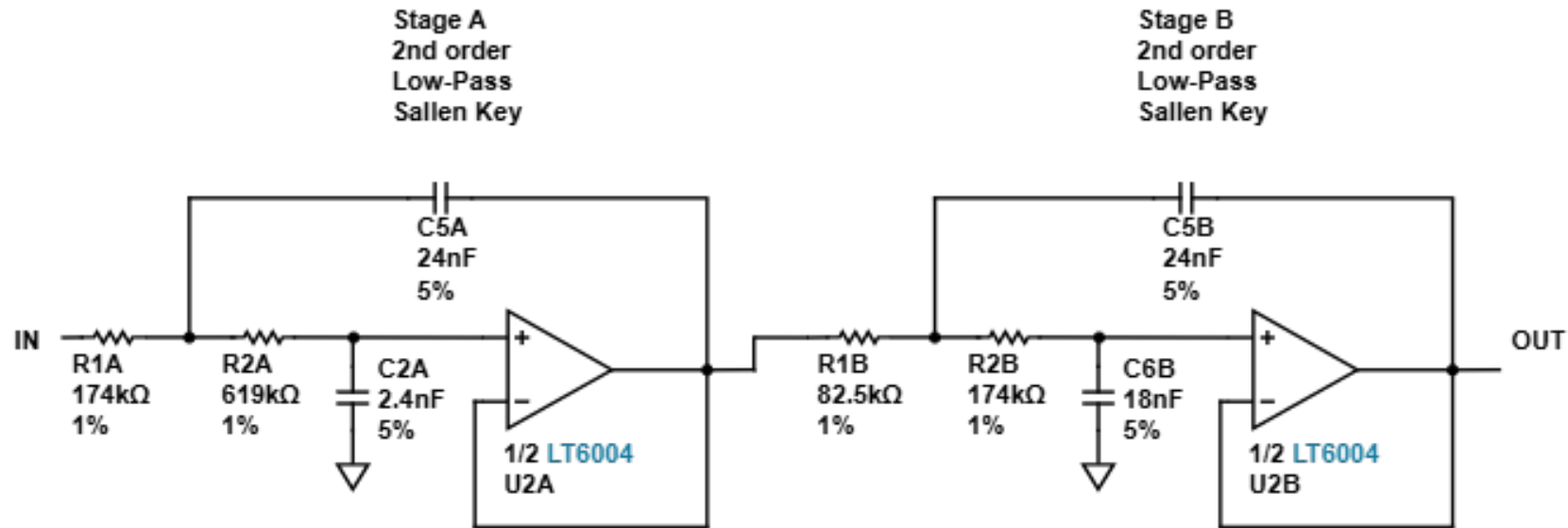
58.7 to 66

541m

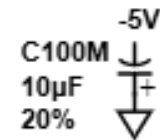
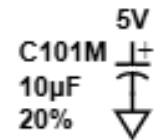
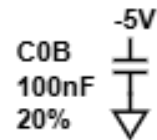
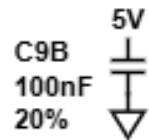
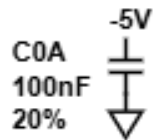
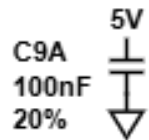
523m to 581m



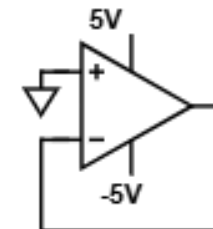
Circuit



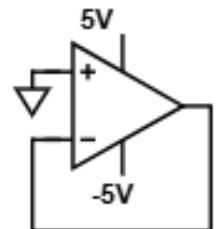
BYPASS CAPACITORS



SPARES Why The Spares?



1/2 LT6004
U2A



1/2 LT6004
U2B